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DATA-DRIVEN STRATEGIES TO IMPROVE RELIABILITY, SAFETY AND RESOURCE ALLOCATION FOR WASHINGTON METROPOLITAN AREA TRANSIT AUTHORITY (WMATA)

FINAL PROJECT- DATABASE MANAGEMENT SYSTEMS

BUDT703_Project_0506_09

MCC SOLUTIONS

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Data Processing, Findings, Insights, and Recommendations

This section details the progression from raw data preparation to actionable business intelligence within the WMATA Performance Analysis project. It traces how data moved through extraction, transformation and loading phases. Analytical processing, strategic interpretation, and visualization demonstrate the database's capacity to support organizational decisions in ridership optimization, service reliability, and safety management.

1.0 Data Sources and Tables

1.1 Data Processing & Preparation

We used data from WMATA's published accounts and reports. Some data was aggregated e.g injury counts, whilst other data originated from siloed operational systems, e.g. faregate logs, bus feeds, performance logs, safety reports, reference files, and calendar systems. In order to use this for forecasting future trends, there was need to extract, transform and load this data as follows:

1.2 Extraction

This purpose of this extraction step was to isolate the data necessary to populate both the dimensional context (Station, BusRoute, CalendarDate) and the performance fact tables (RailPerformance, BusPerformance), ensuring that the analytical environment is comprehensive and aligned with WMATA's business needs.

The raw data found in the published reports was sourced from the primary operational systems that support WMATA's rail and bus activities:

- **Metrorail faregate and station activity logs** – These systems record passenger entries and exits at faregates. To correct for undercounting, these systems also include system-derived estimates for passengers detected passing through without tapping.
- **Metrobus vehicle telematics and operator logs** – Automated vehicle location feeds and operator entries provide boardings, on-time departures, scheduled versus actual trips, and missed trip information.
- **Daily performance reports** – Internal operational reports document escalator availability percentages, reliability indicators, and service exceptions for both rail and bus.
- **Incident reporting systems** – Crime incidents, customer injuries, and station-level safety events are captured and logged through WMATA's incident management tools.

- **Station and route reference files** – Stationary sources provide contextual information for the Station and BusRoute dimensions, including station name, line affiliation, address, and route metadata.
- **Generated calendar data** – The CalendarDate table is sourced through ETL-driven date generation rather than raw operational logs and provides standardized temporal attributes (month, quarter, year, weekday/weekend) required for time-series analytics.

1.3 Cleansing Phase

Cleansing improves the data quality by identifying and correcting errors inherent in the operational sources. Since WMATA's systems are optimized for real-time operations rather than analytical use, the raw data often contains inconsistencies, anomalies, and missing values.

1.3.1 Standardization of Data

It was necessary to standardize data when creating the tables. This involved ensuring the data has the same naming conventions and is of the same data type. Doing this ensures data reliability.

1.3.2 Data Correction and Error Handling

Some data had missing operational metrics and inconsistent station and route names. These had to be standardization and normalized through :

- Using SQL functions like **CAST OR CONVERT** : Ensures that data extracted from different sources are compatible for comparison or manipulation, preventing errors due to data type mismatches during integration
- **Enforcing Entity Integrity :** SQL ensures entity integrity by requiring every relation to have a primary key with non-null components. For clarity, the RailPerformance table implements this through a composite PK of stationID and calendardateID, guaranteeing unique identification of each station's performance record
- **Enforcing Referential Integrity :** Foreign Keys (FKs) are used to ensure that associations between tables are valid, maintaining consistency among rows of related tables. This prevents records (like a performance entry) from existing without a valid reference (like a valid station or date).
- SQL clauses like **ON DELETE NO ACTION** and **ON UPDATE CASCADE** define how the database should handle relational integrity when related data are modified, effectively defining error response rules.
- In the cleansing phase, source system identifiers (such as station names or legacy route codes) were mapped to the standardized primary keys defined in the logical schema i.e stationID, busRouteID, and calendarDateID. This transformation ensured proper referential integrity during subsequent data loading operations.

1.4 Data Transformation using SQL

Data transformation is the process of converting cleansed operational data into the structures, formats, and granularities specified by the target relational schema. This project implements transformation using SQL capabilities including Data Manipulation Language (DML), Data Definition Language (DDL), Common Table Expressions (CTEs), aggregate functions, and conditional logic. The objective is to ensure each fact record aligns with the defined analytical grain (one station/route per day) and that all derived metrics supporting ridership, reliability, and safety analysis are accurately calculated.

1.4.1 Structural Mapping and Normalization

The initial transformation step involves translating the conceptual ERD into a normalized set of relational tables.

Eliminating Composite Attributes

Each relation(row) contains only simple, atomic attributes. Composite attributes from the conceptual model,such as calendarFullDates, are decomposed into individual components during ETL, ensuring the resulting schema adheres to normalization principles and enables efficient query execution.

Mapping Relationships into Associative Entities

Many-to-many relationships identified in the conceptual design (e.g., Station and CalendarDate; BusRoute and CalendarDate) are resolved through associative entities:

- **RailPerformance** : resolves the Station to CalendarDate relationship
- **BusPerformance** : resolves the BusRoute to CalendarDate relationship

These associative entities are implemented as relational tables with composite primary keys formed from the foreign keys of their connected dimensions (e.g., stationID, calendarDateID for railPerformance.

Aggregate Metrics

The primary metric for trend analysis combines passenger entries and exits at faregate locations: SUM (railRidershipEntries + railRidershipExits) As TotalRidership. This calculation supports system-wide performance comparisons, station-level rankings, and temporal trend analysis.

- **Performance rates:** Analytical insights in this project depend primarily on aggregate calculations. SQL functions such as `SUM()`, `AVG()`, `COUNT()` convert granular operational data into actionable performance metrics.

1.5 Loading and Performance Optimization

The final ETL (Extract, Transform, Load) phase ensures transformed data is accurately inserted into the target relational schema while preserving data quality, referential integrity, and query performance. This phase operationalizes the physical design of the database by controlling how data is loaded, validating structural integrity, and configuring performance-enhancing mechanisms such as indexes

1.5.1 Data Placement and Loading Mode

Once the data had been fully cleansed and transformed, it was loaded into the designated tables, namely RailPerformance, BusPerformance, and the supporting dimension tables (Station, BusRoute, CalendarDate). Given the project's focus on historical analysis, the loading process implements an append-only strategy where new operational periods are inserted as additional records without modifying or deleting existing data.

1.5.2 Integrity Enforcement

During the loading phase, the database management system (DBMS) enforces entity integrity and referential integrity constraints, ensuring the resulting database conforms to the logical schema and supports accurate analytical operations.

Entity Integrity

Entity integrity is maintained through NOT NULL constraints applied to all primary key attributes. For the associative fact tables RailPerformance and BusPerformance,these constraints ensure that each composite key (stationID, calendarDateID) or (busRouteID, calendarDateID) uniquely identifies a record with no missing components. This prevents invalid or orphaned fact records from being inserted into the database.

Referential Integrity

Referential integrity is enforced through FOREIGN KEY constraints that link fact tables to their corresponding dimension tables:

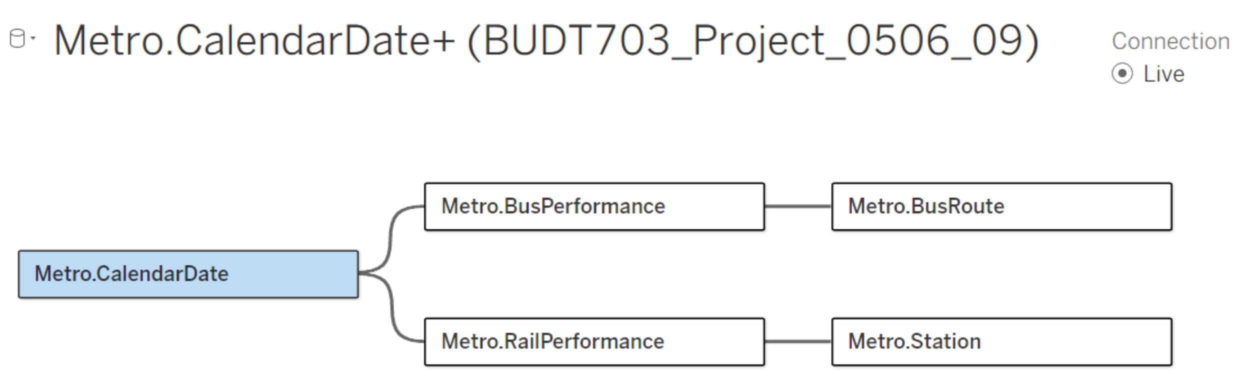
- RailPerformance.stationID linked to Station.**stationID**
- RailPerformance.calendarDateID linked to CalendarDate.**calendarDateID**
- BusPerformance.busRouteID linked to BusRoute.**busRouteID**
- BusPerformance.calendarDateID linked to CalendarDate.**calendarDateID**

These constraints ensure that performance records can only be inserted when their associated station, bus route, and date exist in the respective dimension tables.

2.0 Analytical Methods & SQL Outputs

To address the project’s research questions, a series of descriptive, diagnostic, and time-series analytical methods were implemented using SQL. Each method directly aligns with the mission objectives: analyzing ridership trends, diagnosing performance reliability, identifying safety exposure and resource allocation across Metrorail and Metrobus systems. The queries below represent the core analytical logic used to extract insights from the relational database.

2.1 Completed Data Source (Tableau)



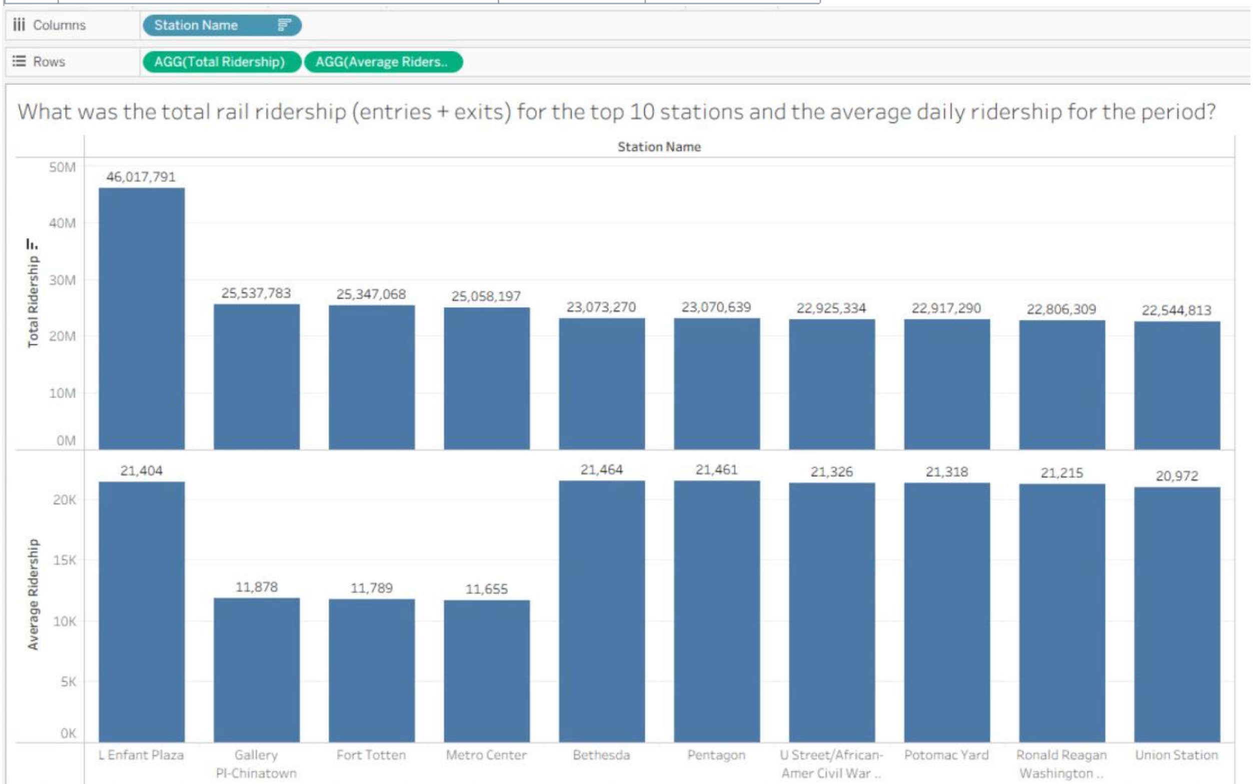
2.2 Objective 1: Analyze Ridership Trends: Track and compare data for the period 2022-2024 relating to ridership patterns for each Metrorail station and Metrobus route, identifying high-growth and low-performing areas.

Business Transaction 1:Average Daily Rail Ridership by Station

Question: What was the total rail ridership (entries + exits) for each station and the average daily ridership for the period? Select Top 10

```
SELECT  TOP 10 s.stationName AS 'Station Name' , SUM(p.railRidershipEntries + p.railRidershipExits)
AS 'Total Ridership',
AVG(p.railRidershipEntries + p.railRidershipExits) AS 'Average Ridership'
FROM [Metro.RailPerformance] p JOIN [Metro.Station] s
    ON p.stationID = s.stationID
GROUP BY s.stationName
ORDER BY [Total Ridership] DESC;
```

	Station Name	Total Ridership	Average Ridership
1	L Enfant Plaza	46017791	21403
2	Gallery Pl-Chinatown	25537783	11878
3	Fort Totten	25347068	11789
4	Metro Center	25058197	11654
5	Bethesda	23073270	21463
6	Pentagon	23070639	21461
7	U Street/African-Amer Civil War Memorial/Cardozo	22925334	21325
8	Potomac Yard	22917290	21318
9	Ronald Reagan Washington National Airport	22806309	21215
10	Union Station	22544813	20971



Insights

- A small subset of stations captures the majority of system ridership.
- L'Enfant Plaza shows approximately 46 million riders over the period.
- This confirms L'Enfant Plaza as the most critical interchange node in the system (Blue, Orange, Silver, Yellow, Green connectivity).
- This station has both the highest operational demand and highest exposure to crowding-related safety risks.
- High-traffic stations typically serve as multimodal hubs or employment centers.

Recommendations

- Focus operational resources (staffing, cleaning, crowd control) on the top 10 stations.
- Schedule maintenance during off-peak periods to minimize rider impact.
- Prioritize these locations for capacity upgrades and accessibility enhancements.

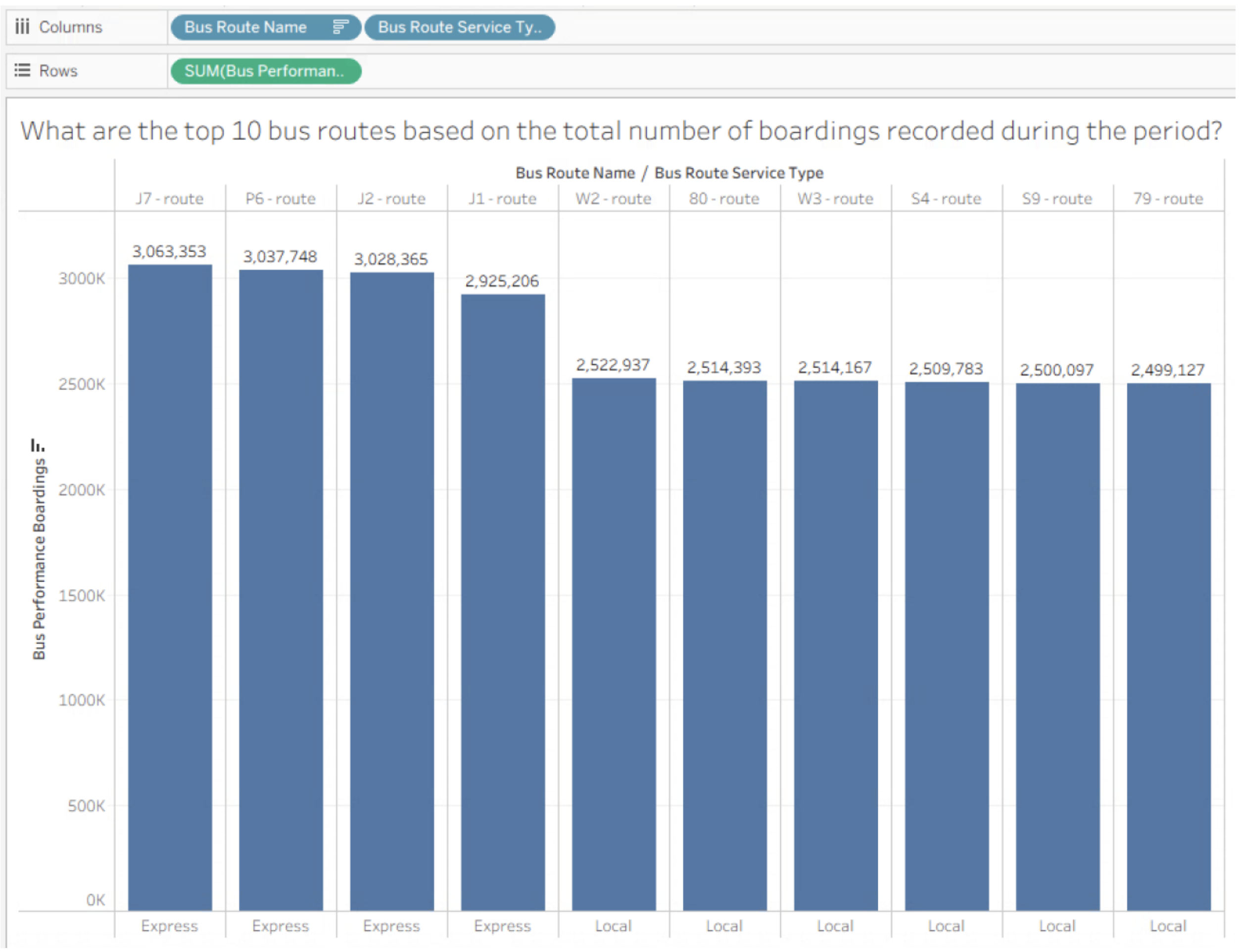
- These findings underscore the need for WMATA to prioritize resource allocation, staffing, and capital improvements at the busiest stations to support reliability, safety, and customer experience

Business Transaction 2: Top Bus Routes by Ridership

Question: What are the top 10 bus routes based on the total number of boardings recorded during the period?

```sql
SELECT TOP 10 r.busRouteName AS 'Bus Route', r.busRouteServiceType AS 'Route Service Type',
SUM(b.busPerformanceBoardings) AS 'Total Boardings'
FROM  [Metro.BusPerformance] b JOIN [Metro.BusRoute] r
    ON b.busRouteID = r.busRouteID
GROUP BY r.busRouteID, r.busRouteName, r.busRouteServiceType
ORDER BY [Total Boardings] DESC;
```

	Bus Route	Route Service Type	Total Boardings
1	J7 - route	Express	3063353
2	P6 - route	Express	3037748
3	J2 - route	Express	3028365
4	J1 - route	Express	2925206
5	W2 - route	Local	2522937
6	80 - route	Local	2514393
7	W3 - route	Local	2514167
8	S4 - route	Local	2509783
9	S9 - route	Local	2500097
10	79 - route	Local	2499127



Insights

- The bus network exhibits significant demand concentration among a limited number of high-performing routes.
- These routes predominantly serve essential connectivity functions between residential communities and employment centers or major transit interchange points.

Recommendations

- Direct operational priority to high-volume routes by enhancing service frequency, strengthening preventive maintenance schedules, and deploying senior operators.
- Integrate ridership pattern analysis into route network planning processes and fleet allocation optimization frameworks.

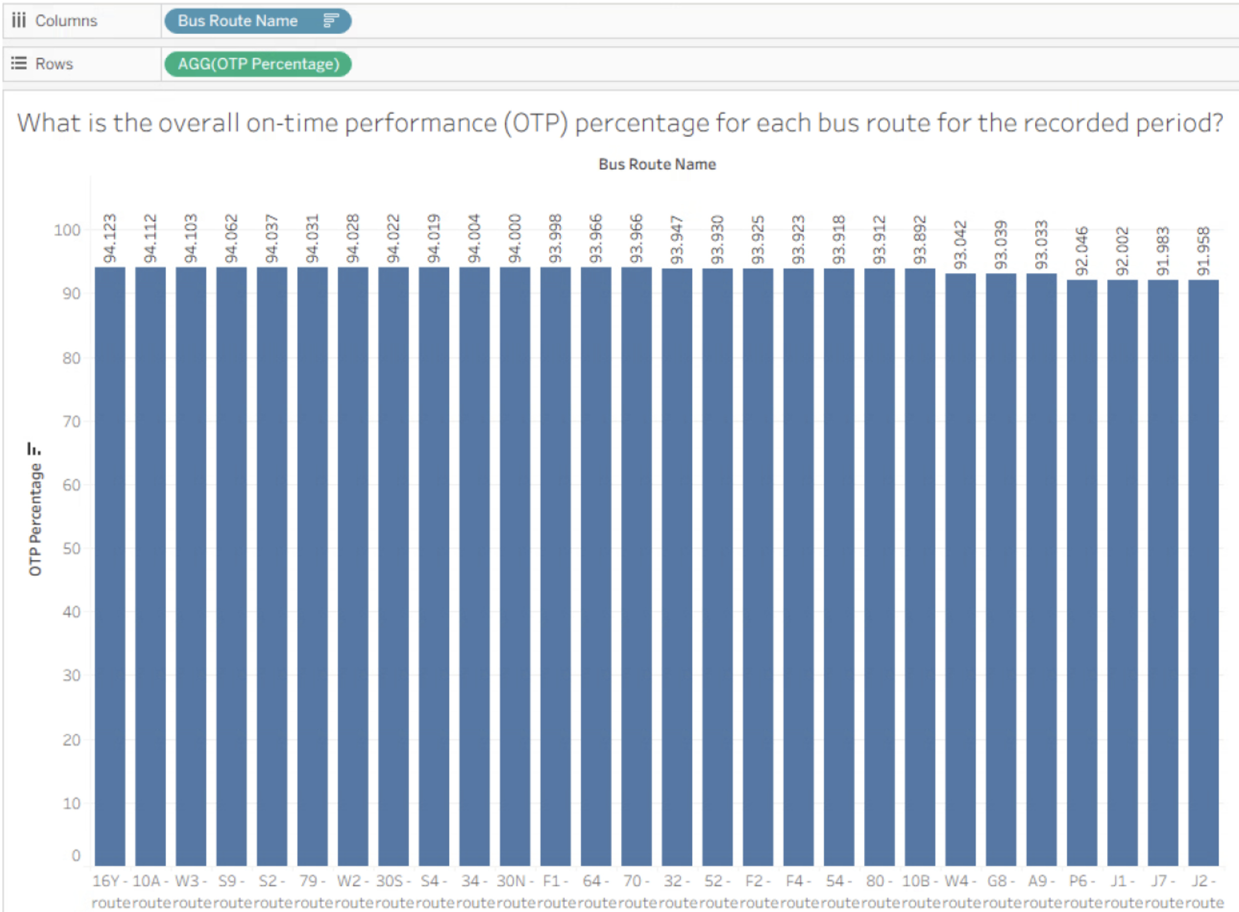
2.3 Objective 2: Correlate service performance with reliability: Evaluate operational metrics such as on-time departures to determine how reliability affects customer experience and ridership.

Business Transaction 3 : On-Time Performance by Bus Route

Question: What is the overall on-time performance (OTP) percentage for each bus route for the recorded period?

```sql
SELECT r.busRouteName AS 'Bus Route', (SUM(CAST(b.busPerformanceTimeDepartures AS DECIMAL(18,6)))
    / NULLIF(SUM(CAST(b.busPerformanceTotalDepartures AS DECIMAL(18,6))),0)) * 100
      AS 'OTP Percentage'
FROM [Metro.BusRoute] r JOIN [Metro.BusPerformance] b
    ON b.busRouteID = r.busRouteID
GROUP BY r.busRouteID, r.busRouteName
ORDER BY [OTP Percentage] DESC;
```

	Bus Route	OTP Percentage
1	16Y - route	94.123400
2	10A - route	94.111700
3	W3 - route	94.103300
4	S9 - route	94.062300
5	S2 - route	94.036800
6	79 - route	94.031100
7	W2 - route	94.027600
8	30S - route	94.021800
9	S4 - route	94.018900
10	34 - route	94.003900
11	30N - route	93.999800
12	F1 - route	93.997500
13	64 - route	93.966100
14	70 - route	93.966100
15	32 - route	93.946600
16	52 - route	93.930000
17	F2 - route	93.925100
18	F4 - route	93.922900
19	54 - route	93.918000
20	80 - route	93.912100
21	10B - route	93.891900
22	W4 - route	93.041500
23	G8 - route	93.039300
24	A9 - route	93.033000
25	P6 - route	92.046400
26	J1 - route	92.001500
27	J7 - route	91.983000
28	J2 - route	91.958000



Insights

- The OTP is tightly clustered around approximately 91%-94%
- There are very few major outliers. This indicates a high degree of reliability and standardization across the routes shown.
- The uniformity of the data makes ranking them uninformative.

Recommendations

- Given that WMATA's system-wide targets typically ranged around 77% in FY24, this may mean that this data is cleaner than WMATA's data.
- It may be necessary to obtain raw data to reveal the true reliability issues of ridership impact.

2.3.1 **Business Transaction 4:** Incident vs. Ridership Correlation

Question: What is the correlation between top 10 stations with highest incidents against stations highest?

```
WITH IncidentRank AS (
    SELECT TOP 10 s.stationID,s.stationName AS 'Station Name',s.stationJurisdiction AS 'Station Jurisdiction',
    SUM(r.railCrimeIncidentCount + r.railCustomerInjuryCount) AS 'Total Incidents',
    RANK() OVER (ORDER BY SUM(r.railCrimeIncidentCount + r.railCustomerInjuryCount) DESC) AS 'Incident Rank'
    FROM [Metro.Station] s
    JOIN [Metro.RailPerformance] r ON s.stationID = r.stationID
    GROUP BY s.stationID, s.stationName, s.stationJurisdiction),RidershipRank AS (
                SELECT  s.stationID,s.stationName,
        SUM(r.railRidershipEntries + r.railRidershipExits) AS 'Total Ridership',
        RANK() OVER (ORDER BY SUM(r.railRidershipEntries + r.railRidershipExits) DESC) AS 'Ridership Rank'
                FROM [Metro.Station] s
                JOIN [Metro.RailPerformance] r  ON s.stationID = r.stationID
        GROUP BY s.stationID, s.stationName)
                        SELECT [Station Name],[Station Jurisdiction],[Total Incidents],[Incident Rank],[Total Ridership], [Ridership Rank], ([Ridership Rank] - [Incident Rank]) AS 'Rank Difference'
                        FROM IncidentRank i JOIN RidershipRank r ON i.stationID = r.stationID
ORDER BY [Incident Rank];
```

	Station Name	Station Jurisdiction	Total Incidents	Incident Rank	Total Ridership	Ridership Rank	Rank Difference
1	L Enfant Plaza	Washington DC	113	1	23112043	2	1
2	U Street/African-Amer Civil War Memorial/Cardozo	Washington DC	103	2	22925334	6	4
3	Metro Center	Washington DC	102	3	22795384	10	7
4	Gallery Pl-Chinatown	Washington DC	102	3	23280694	1	-2
5	Rosslyn	Arlington VA	102	3	22385274	12	9
6	Ronald Reagan Washington National Airport	Arlington VA	101	6	22806309	9	3
7	Union Station	Washington DC	101	6	22544813	11	5
8	Bethesda	Bethesda MD	100	8	23073270	4	-4
9	Pentagon	Arlington VA	100	8	23070639	5	-3
10	Potomac Yard	Alexandria VA	99	10	22917290	7	-3

Insights

- Safety incident patterns reveal significant deviations from ridership-based expectations: some stations report incident levels that substantially exceed their passenger traffic rankings.
- Other stations successfully maintain minimal incident rates while serving high passenger volumes, indicating achievable safety standards under demanding operational conditions.
- This non-correlation between ridership and incidents demonstrates that safety risk originates from station-specific environmental conditions, design characteristics, and operational protocols rather than passenger density alone.

Recommendations

- Allocate investigative priority to stations exhibiting positive RankDifference anomalies (where incident rankings substantially exceed ridership rankings).
- Deploy comprehensive risk mitigation measures including enhanced uniformed presence, reconfigured passenger circulation systems to reduce conflict points, and physical security improvements through upgraded illumination and video monitoring systems.

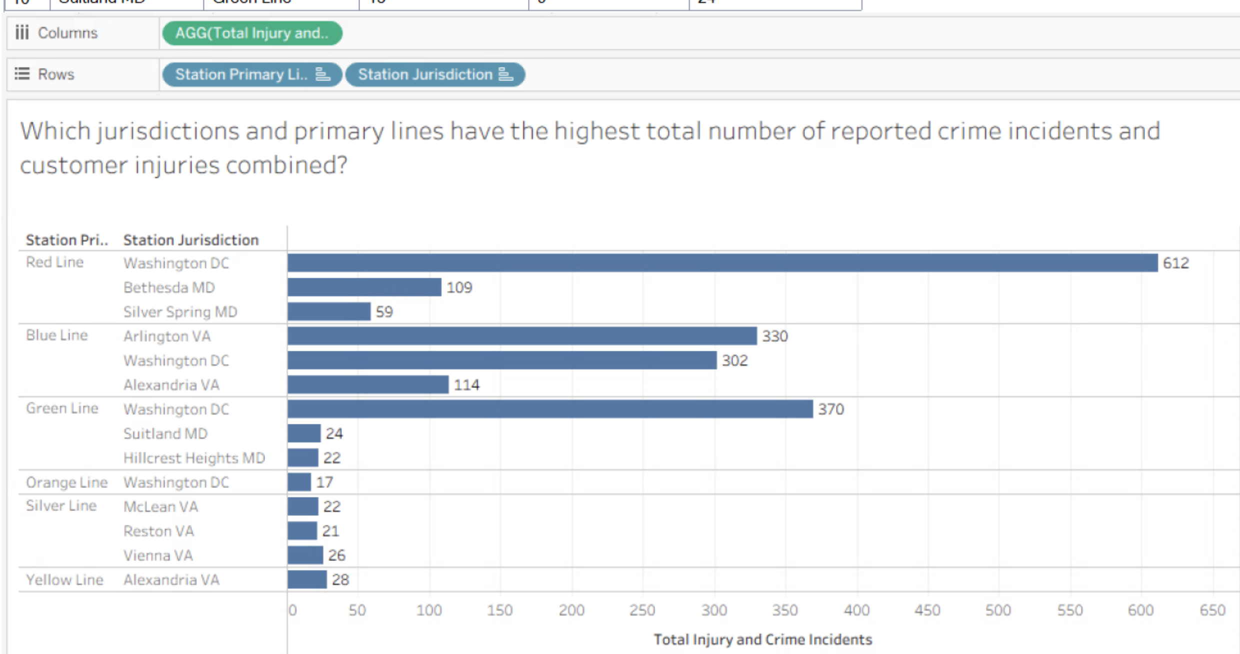
- Systematically document and replicate operational excellence from positive outlier stations, those demonstrating strong safety performance despite high passenger volumes,to establish system-wide best practices.

2.4 Objective 3: Enhance safety and security insights: Identify and map safety incidents by Metrorail station and line.

Business Transaction 5 : Safety Incidents by Jurisdiction and Line
Question: Which Top 10 jurisdictions and primary lines have the highest total number of reported crime incidents and customer injuries combined?

```
SELECT TOP 10 s.stationJurisdiction AS 'Station Jurisdiction',s.stationPrimaryLine
AS 'Station PrimaryLine' , SUM(r.railCrimeIncidentCount) AS TotalReportedCrimes,SUM
(r.railCustomerInjuryCount) AS TotalCustomerInjury,
SUM(r.railCrimeIncidentCount + r.railCustomerInjuryCount) AS 'Total Safety
Incidents'
FROM [Metro.Station] s JOIN [Metro.RailPerformance] r
    ON  s.stationID = r.stationID
GROUP BY s.stationJurisdiction , s.stationPrimaryLine
ORDER BY [Total Safety Incidents] DESC;
```

	Station Jurisdiction	Station PrimaryLine	TotalReportedCrimes	TotalCustomerInjury	Total Safety Incidents
1	Washington DC	Red Line	348	264	612
2	Washington DC	Green Line	201	169	370
3	Arlington VA	Blue Line	179	151	330
4	Washington DC	Blue Line	161	141	302
5	Alexandria VA	Blue Line	63	51	114
6	Bethesda MD	Red Line	61	48	109
7	Silver Spring MD	Red Line	29	30	59
8	Alexandria VA	Yellow Line	19	9	28
9	Vienna VA	Silver Line	13	13	26
10	Suitland MD	Green Line	15	9	24



2.5 Objective 4: Support Performance Ranking and Resource Allocation: Enable ranking of stations and routes by ridership, reliability, and safety metrics to inform planning, maintenance prioritization, and service redesign initiatives.

Business Transaction 6 : Crime, Injuries, and Ridership by Station
Question: Which rail stations had the highest total number of crime incidents and customer injuries, and what was the total ridership for those stations?

```
SELECT TOP 10 s.stationName AS 'Station Name',s.stationJurisdiction 'Station Jurisdiction',
SUM(p.railCrimeIncidentCount)  AS 'Count of Crime Incidents', SUM(p.railCustomerInjuryCount) AS
'Count of Customer Injuries',
 SUM(p.railRidershipEntries + p.railRidershipExits) AS 'Total Ridership'
FROM [Metro.Station] s JOIN [Metro.RailPerformance] p
    ON s.stationID = p.stationID
GROUP BY s.stationName, s.stationJurisdiction
ORDER BY [Count of Crime Incidents] DESC, [Count of Customer Injuries] DESC;
```

	Station Name	Station Jurisdiction	Count of Crime Incidents	Count of Customer Injuries	Total Ridership
1	L Enfant Plaza	Washington DC	104	103	46017791
2	Metro Center	Washington DC	69	42	25058197
3	Gallery Pl-Chinatown	Washington DC	67	49	25537783
4	Ronald Reagan Washington National Airport	Arlington VA	60	41	22806309
5	Rosslyn	Arlington VA	59	43	22385274
6	Bethesda	Bethesda MD	57	43	23073270
7	Fort Totten	Washington DC	57	43	25347068
8	Potomac Yard	Alexandria VA	54	45	22917290
9	Union Station	Washington DC	51	50	22544813
10	Pentagon	Arlington VA	50	50	23070639

Insights

- L'Enfant Plaza has the highest safety incidents.
- The gap between L'Enfant Plaza is significant, showing high safety risk.
- Incident frequency generally tracks with ridership volume across the station network as shown on major hubs such as Metro Centre and Gallery Pl.
- Though important exceptions such as Bethesda and Fort Totten reveal that ridership alone does not fully explain safety performance. These areas have higher than expected safety metrics, despite high traffic.
- This supports the insight that high incidence is not always a function of high ridership.

Recommendations

- Institute targeted infrastructure evaluations at high-incident stations such as L'Enfant Plaza, with specialized focus on escalator/elevator systems and platform capacity constraints.
- Expand operational supervision through dedicated customer assistance and crowd management personnel during peak service periods.
- Improve patrol, lighting , signage and patrol in high safety risk areas.
- Identify other factors that are not associated with traffic volume, that may influence safety risks eg traffic flow problems.

3. Limitations of the Study

This study provides important insights into WMATA's operational performance, however, several limitations affect result precision and completeness. These stem from source data constraints, ETL methodology choices, and general complexities in integrating various operational systems.

3.1 Data Source Limitations

- Incomplete Metrobus Ridership Fare data because of fare-evasion causes systematic underreporting of bus ridership, affecting demand and reliability analyses. This structural limitation cannot be resolved without improved validation compliance .
- The estimated Metrorail Ridership Faregate counts include algorithmic "no-tap" estimates. This brings some inherent measurement of error in the data collected.
- Historical data originates from disconnected systems with inconsistent naming, granularity, and completeness. Despite ETL reconciliation efforts, residual inconsistencies may persist.
- Implementation of a unified database system requires extensive backup and recovery procedures.

- WMATA's calculation of OTP is not as narrow as the one used in the project. WMATA includes thresholds (e.g. 2 mins early or late). This may account for the higher OTP in this dataset.

References

Books

Hoffer, J. A., Ramesh, V., & Topi, H. (2016). *Modern database management* (12th ed., Global Edition). Pearson Education Limited.

WMATA Published Reports and Publications

Washington Metropolitan Area Transit Authority. (2024, October). *Network redesign final summary report.*

Washington Metropolitan Area Transit Authority. (2025, June). *FY25 Q3 service excellence report.*

Washington Metropolitan Area Transit Authority. (n.d.). *Ridership data portal.* https://www.wmata.com

Washington Metropolitan Area Transit Authority. (n.d.). *Notice of public hearing: Proposed FY2025 Capital Improvement Program.*

Washington Metrorail Safety Commission. (2023). *Annual report on the safety of the WMATA rail system in 2023.*

Lecture Notes

Lee, A. (2025). *Database Management Systems: Lecture notes*